Towards Practical Ultrasound-Based Hand Motion Analysis: Efficient Reference Acquisition and Motion Modeling

Christian Kupsch[1], Nico Schurig[1] Robert Kunz[1], Lennart Schierling[1], David Weik[1]
[1]Measurement, Sensor and Embedded Systems Laboratory (MSE Lab), TU Bergakademie Freiberg, Freiberg, Germany

Background, Motivation and Objective

Ultrasound-based sensing at the forearm is a promising approach for hand motion analysis and future human-machine interaction systems, as it can capture internal anatomical changes related to muscle and tendon activity. However, practical adoption remains challenging due to the effort required for calibration, the complexity of suitable motion representations, and the computational demands of data-driven processing. The objective of this work is to develop methods that support a practical workflow for ultrasound-based hand motion analysis and prepare the path towards portable and embedded implementations.

Statement of Contribution/Methods

For a practical calibration, our current work demonstrates how a standard webcam can be used for an efficient generation of motion-labeled reference data. Ultrasound images are captured from the forearm muscles, whilst the webcam records the hand motion which is traced using MediaPipe to allow for automated extraction of hand landmarks from video data. For efficient modeling, hand and finger motion models are investigated to transform high-dimensional landmark information into more compact and anatomically meaningful target variables. Furthermore, embedded deployment of machine learning models in potentially wearable systems is pursued by hardware and ressource aware neural architecture search in ongoing work.

Results/Discussion

Initial results demonstrate the feasibility of deriving motion-related hand information from forearm ultrasound data using the proposed reference acquisition and modeling strategy. In particular, an efficient motion model based on relative joint angles allows the learning problem to be formulated in a more compact and physiologically plausible way. The presented approach reduces the barrier for experimental calibration and establishes a flexible methodological basis for future ultrasound wearable systems. Beyond current stationary setups, the combination of efficient reference acquisition, reduced motion modeling, and embedded machine learning opens promising perspectives for practical real-time applications.